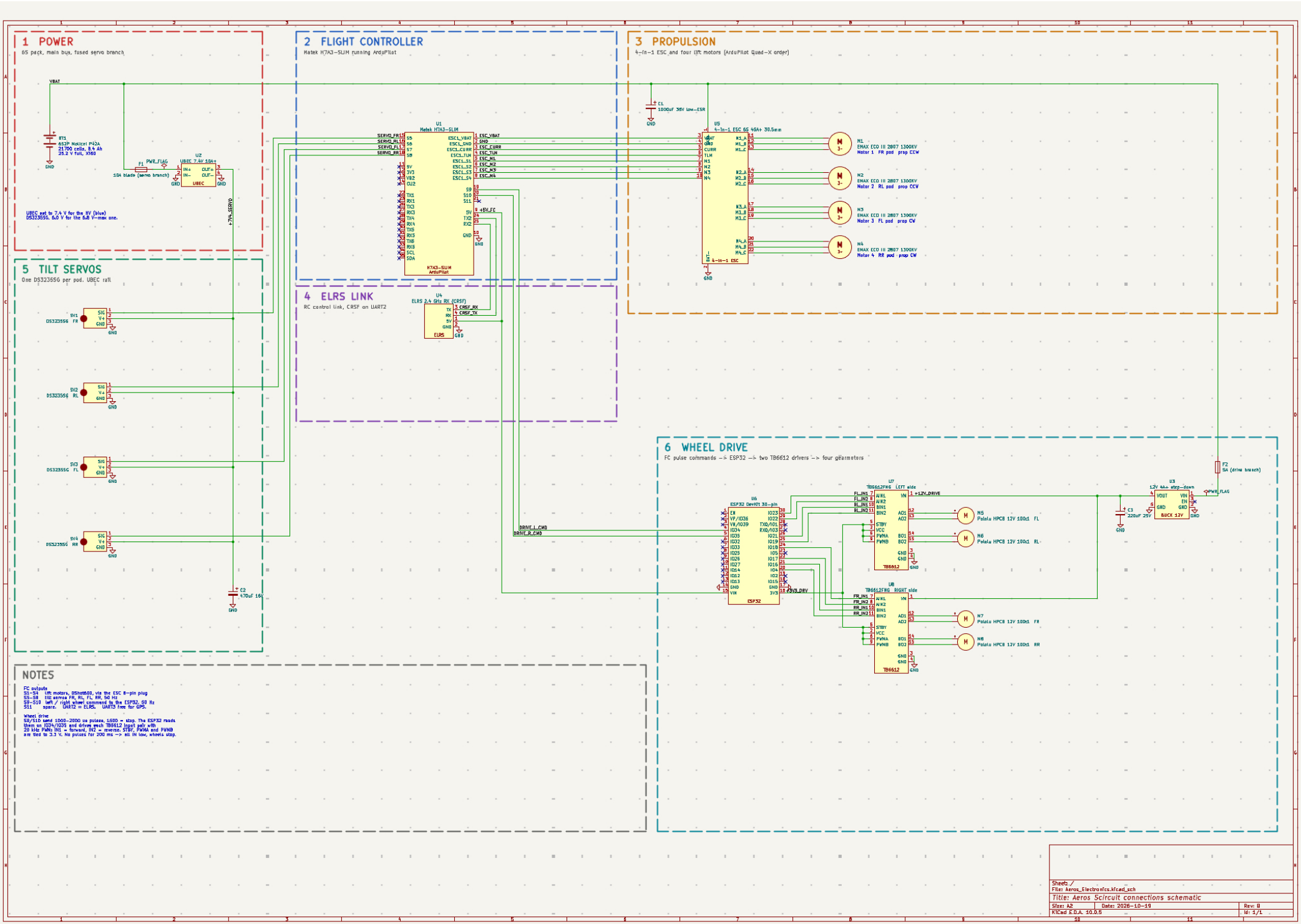

1 POWER
6S pack, main bus, fused servo branch
VBAT
BT1
6S2P Molicel P42A
21700 cells, 8.4 Ah
25.2 V full, XT60
GND
F1
PWR_FLAG
10A blade (servo branch)
U2
UBEC 7.4V 10A+
IN+
IN-
OUT+
OUT-
UBEC
UBEC set to 7.4 V for the RV (blue)
DS3235SG. 6.0 V for the 6.8 V-max one.
+7.4V_SERVO
2 FLIGHT CONTROLLER
Matek H7A3-SLIM running ArduPilot
U1
Matek H7A3-SLIM
H7A3-SLIM
ArduPilot
SERVO_FR
SERVO_RL
SERVO_FL
SERVO_RR
ESC1_VBAT
ESC1_GND
ESC1_CURR
ESC1_TLM
ESC1_S1
ESC1_S2
ESC1_S3
ESC1_S4
+6V_FC
3 PROPULSION
4-in-1 ESC and four lift motors (ArduPilot Quad-X order)
C1
1000uF 35V low-ESR
U5
4-in-1 ESC 65 45A+ 30.5mm
4-in-1 ESC
M1
EMAX ECO III 2807 1300KV
Motor 1 FR pod prop CCW
M2
EMAX ECO III 2807 1300KV
Motor 2 RL pod prop CCW
M3
EMAX ECO III 2807 1300KV
Motor 3 FL pod prop CW
M4
EMAX ECO III 2807 1300KV
Motor 4 RR pod prop CW
4 ELRS LINK
RC control link, CRSF on UART2
U4
ELRS 2.4 GHz RX (CRSF)
ELRS
CRSF_RX
CRSF_TX
5 TILT SERVOS
One DS3235SG per pod. UBEC rail
SV1
DS3235SG FR
SV2
DS3235SG RL
SV3
DS3235SG FL
SV4
DS3235SG RR
SIG
V+
GND
C2
470uF 16V
6 WHEEL DRIVE
FC pulse commands -> ESP32 -> two TB6612 drivers -> four gearmotors
U6
ESP32 DevKit 30-pin
ESP32
DRIVE_L_CMD
DRIVE_R_CMD
+3V3_DEV
U7
TB6612FNG LEFT side
TB6612
+12V_DRIVE
M5
Pololu HPCB 12V 100:1 FL
M6
Pololu HPCB 12V 100:1 RL
U8
TB6612FNG RIGHT side
M7
Pololu HPCB 12V 100:1 FR
M8
Pololu HPCB 12V 100:1 RR
U3
12V 4A+ step-down
BUCK 12V
C3
220uF 25V
F2
5A (drive branch)
PWR_FLAG
NOTES
FC outputs
S1–S4 lift motors, DShot600, via the ESC 8-pin plug
S5–S8 tilt servos FR, RL, FL, RR, 50 Hz
S9–S10 left / right wheel command by the ESP32, 50 Hz
S11 spare. UART2 = ELRS. UART3 free for GPS.
Wheel drive
S9/S10 send 1000–2000 us pulses, 1500 = stop. The ESP32 reads them on IO34/IO35 and drives each TB6612 input pair with 20 kHz PWMs IN1 = forward, IN2 = reverse, STBY, PWMA and PWMB are tied to 3.3 V. No pulses for 200 ms -> all IN low, wheels stop.
Sheets /
File: Aeros_Electronics.kicad_sch
Title: Aeros Scircuit connections schematic
Size: A2
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Rev: B
KiCad E.D.A. 10.0.5
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